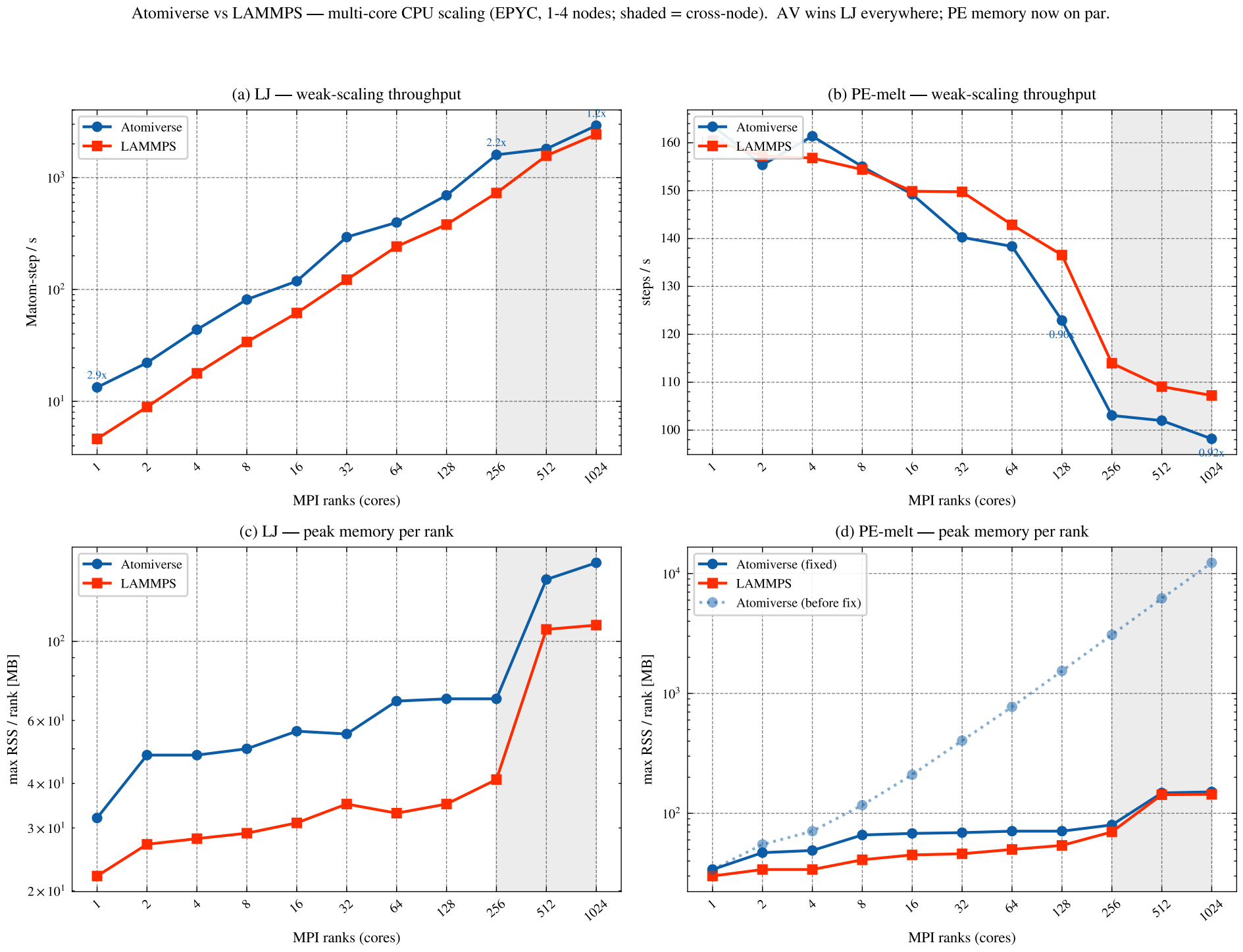
Atomiverse vs LAMMPS — multi-core CPU scaling (EPYC, 1-4 nodes; shaded = cross-node). AV wins LJ everywhere; PE memory now on par.
(a) LJ — weak-scaling throughput
Atomiverse
LAMMPS
Matom-step / s
MPI ranks (cores)
2.9x
2.2x
1.2x
(b) PE-melt — weak-scaling throughput
steps / s
0.90x
0.92x
(c) LJ — peak memory per rank
max RSS / rank [MB]
(d) PE-melt — peak memory per rank
Atomiverse (fixed)
Atomiverse (before fix)